Systems Analysis and Control

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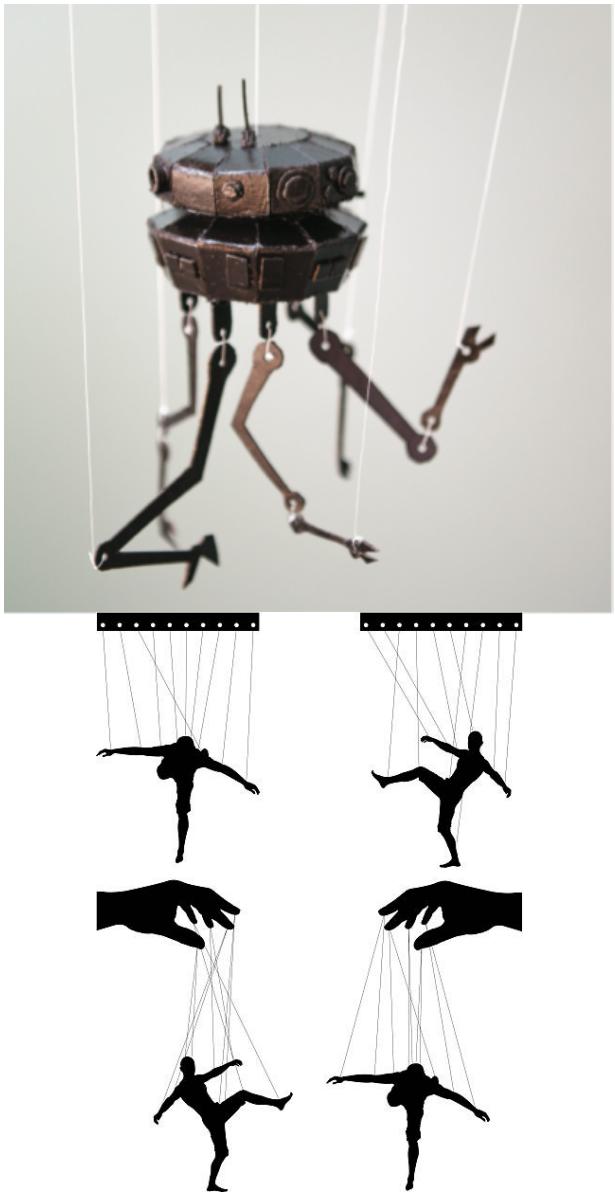
Arizona State University

Lecture 1: Introduction to Control Systems and Historical Perspective

Rorschach Test for Control? Which One is Different?



Star Wars Marionette

Lt. Cmdr. Data from Star Trek TNG

Atlas Robot from Boston Dynamics

What do we mean by "Control"?

Control is the study of how to make things do what you want.

The most robust **Controller** is **Human Consciousness**.

- Fly an airplane.
- Drive a car without crashing.

The Subconscious is also not bad

- Keeping your eyes on the instructor.
- Keeping your heart beating.
- Standing up.

In this class we focus on *Automatic Control*.

- Machines are clay. To function, they must be controlled
  - They only speak **Math**.
- Your subconscious is very good at **Math**.

Unfortunately, you conscious mind is NOT.

Who Am I?

Website: http://control.asu.edu

Research Faculty: Computation, Optimization and Control

Applications:

- Nuclear Fusion Energy
- Spacecraft Navigation
- Communication Networks
- Immunology

Theory:

- Control of Delayed Systems
- Control of PDE Systems
- Control of Nonlinear Systems
- Optimization/AI/Machine Learning

My Background:

- B.Sc. Aerospace Engineering and B.Sc. Physics, University of Texas at Austin
- Ph.D. Aeronautics and Astronautics, Stanford University
- Worked at ARL(GPS), NASA Johnston, Gravity Probe B, and SNAP

Office: ERC 253; Lab: GWC 531

The Difference between **Signals** and **Systems**?

What is a System?

- AKA: "process", "machine", "plant", "thing", et c.
- Can be a "real" thing or a mathematical representation (model).

Definition 1.

A **System** is anything with Inputs and Outputs.
The Inputs and Outputs are **Signals** (information).

There should ALWAYS be Inputs and Outputs!

- **If No Inputs**: You can't change anything.
- **IF No Outputs**: Then it doesn't matter anyway.

Let start with some examples.

Example: A Stereo Receiver

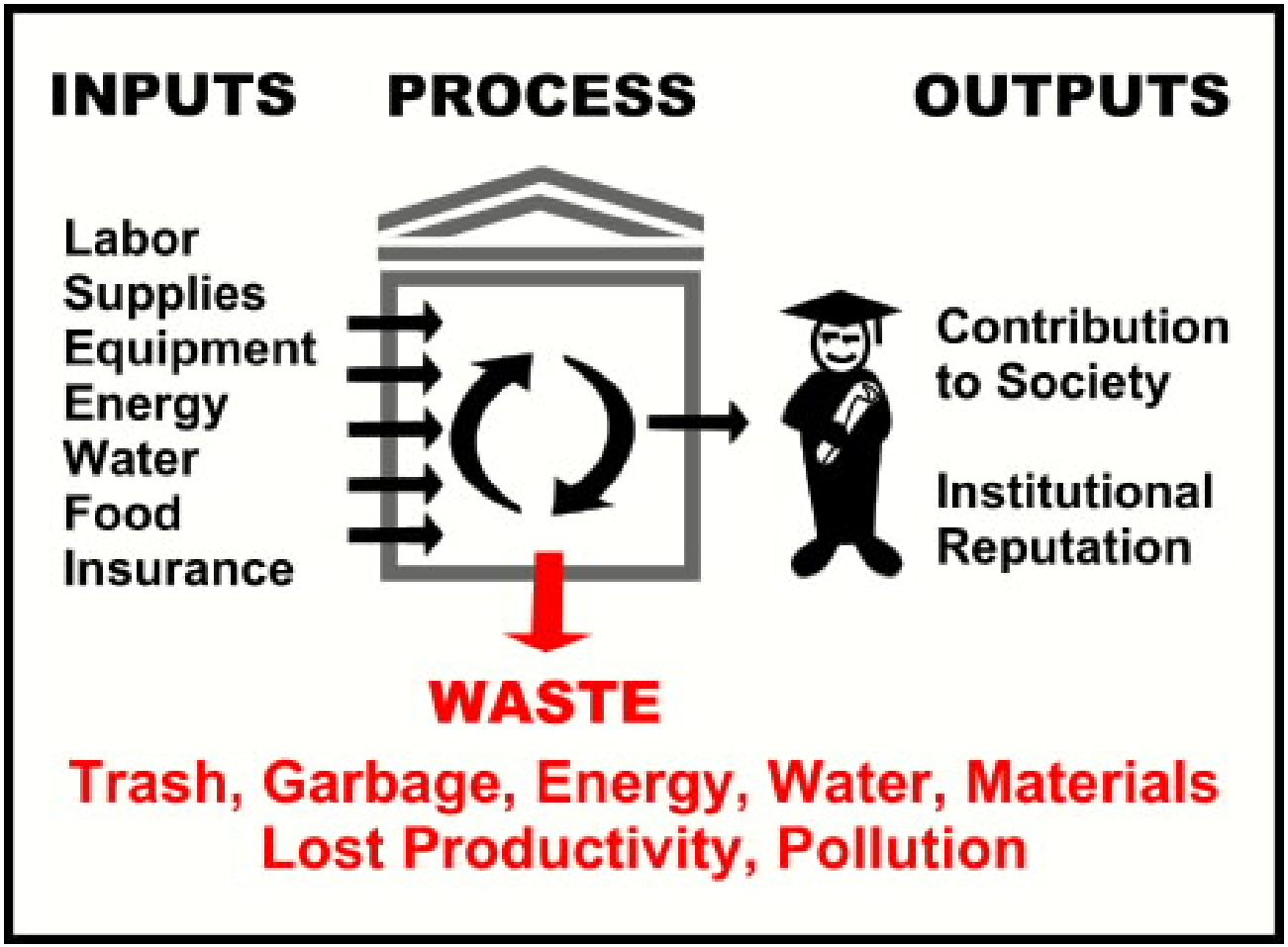
Inputs and Outputs depend on what we are trying to do.

- A system may have multiple Inputs and Outputs.
- But only some of these are relevant to you.

Example of a System: Education

Definition 2.

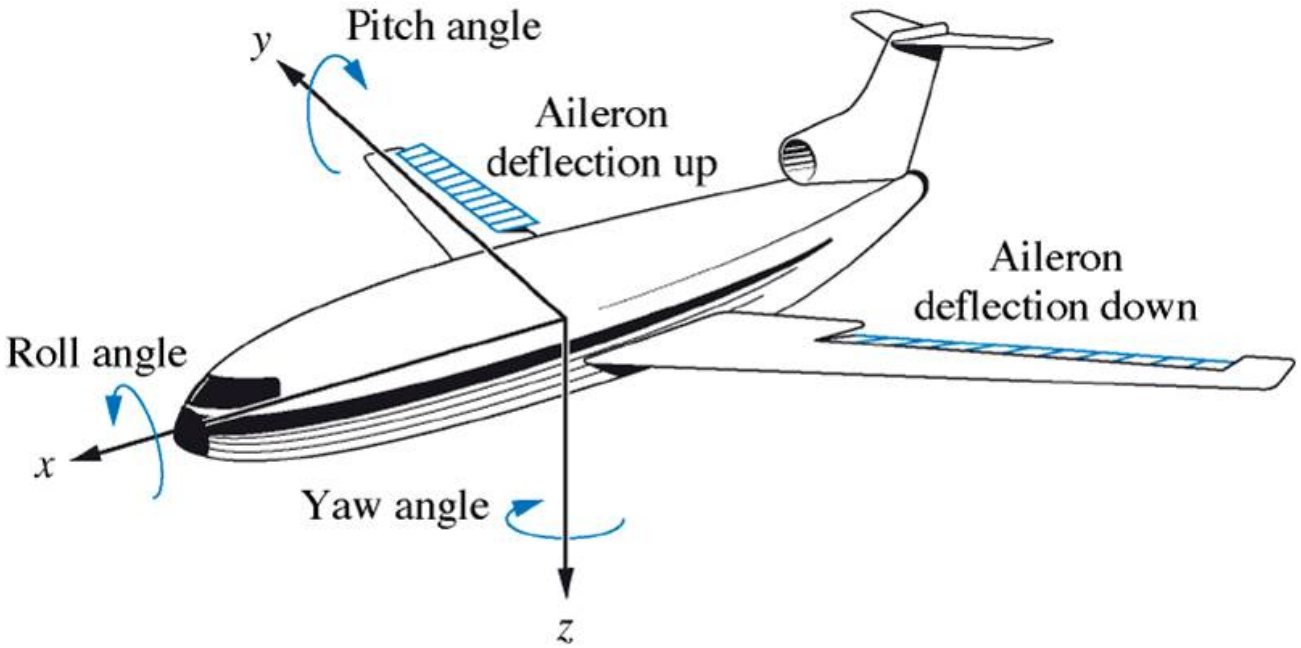
The System to be controlled is called the **Plant**.



WASTE is ALSO an Output!

Example of a System

Aircraft



Inputs: Actuators

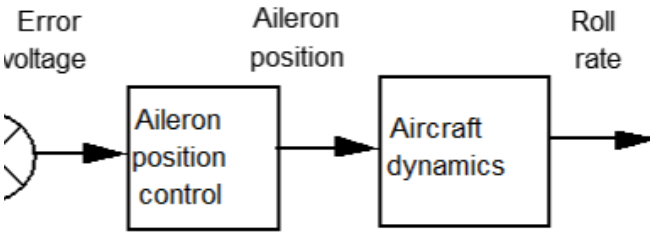
Inputs can be created by **Actuators**.

Definition 3.

An **Actuator** is any *controllable* mechanism/signal/communication Input which can affect the Outputs of the plant.

Examples:

- Ailerons, Rudder
- Force Transducers: Servos/Motors
  - Robots
  - Engines



Actuators are also systems with inputs and outputs. For an aileron:

- Input: Voltage
- Output: Torque

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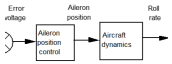

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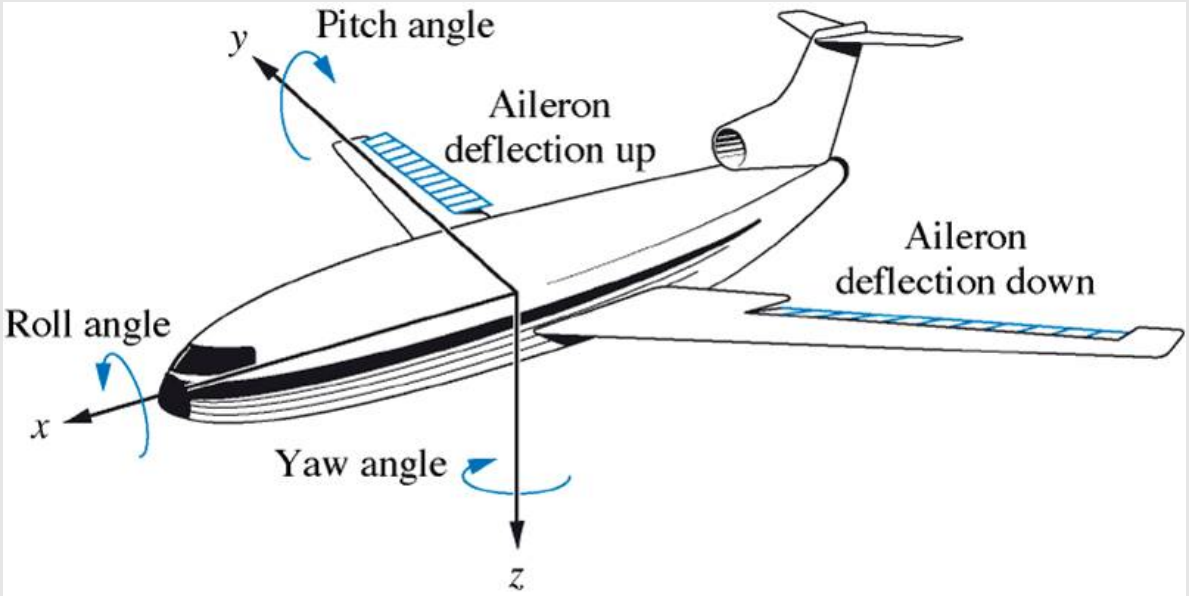
Definition 4.

A **Disturbance** is any *uncontrollable* mechanism/signal/communication Input which can affect the Outputs of the plant.



Examples:

- Wind gusts
- vibrations
- electrical noise

Outputs: Sensors, Measurements and Feedback

Outputs are measured by **Sensors**.

Definition 5.

A **Sensor** is any mechanism which can be used to *measure* an Output (Signal).

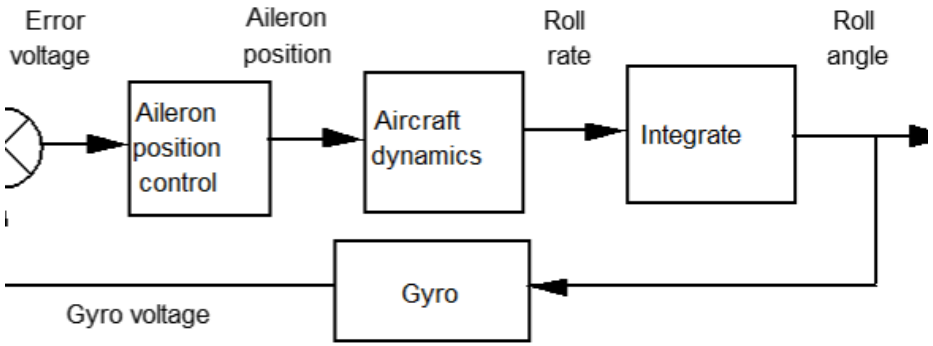
The **PLANT** maps the Inputs to the Outputs.

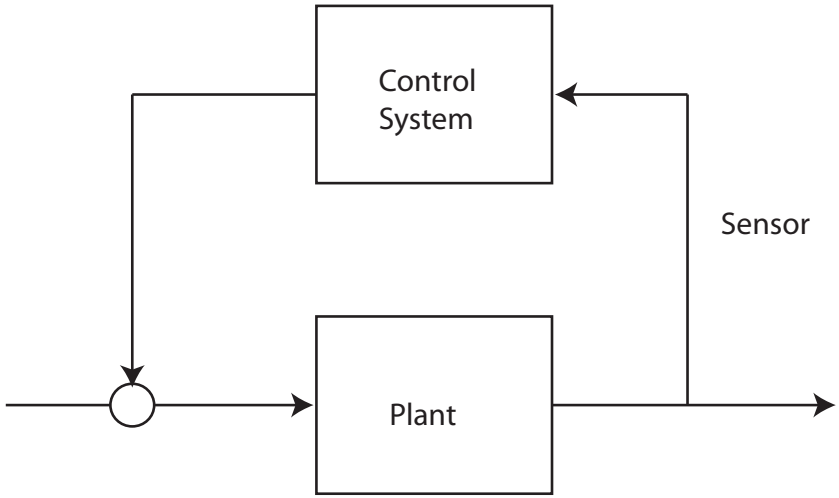
Definition 6.

FEEDBACK maps the Outputs back to the Inputs.
Sensors measure Output and *Actuators* alter the Input.

More Examples of Feedback:

- Aircraft Pitch Control





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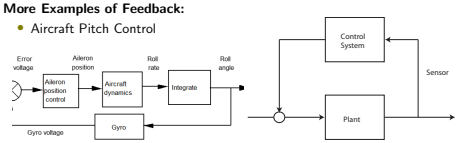
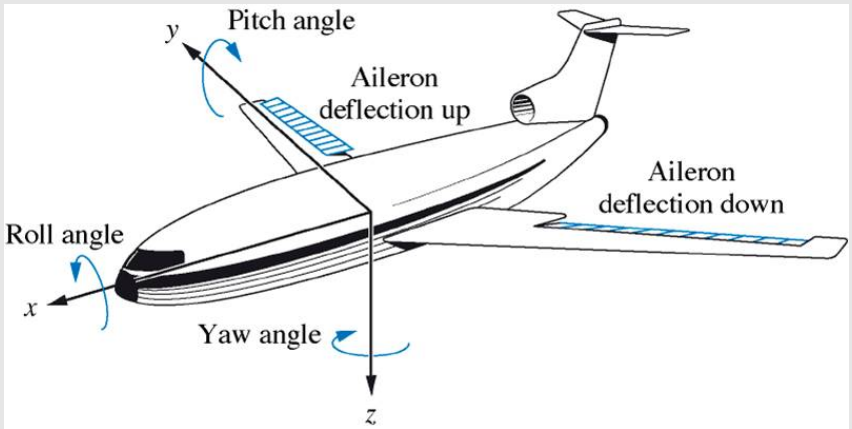


Definition 7.

A **Regulated Output** is any Output which is of practical interest.

- **Regulated Outputs** may not necessarily be measurable using Sensors
- **Measured Outputs** may not necessarily be regulated

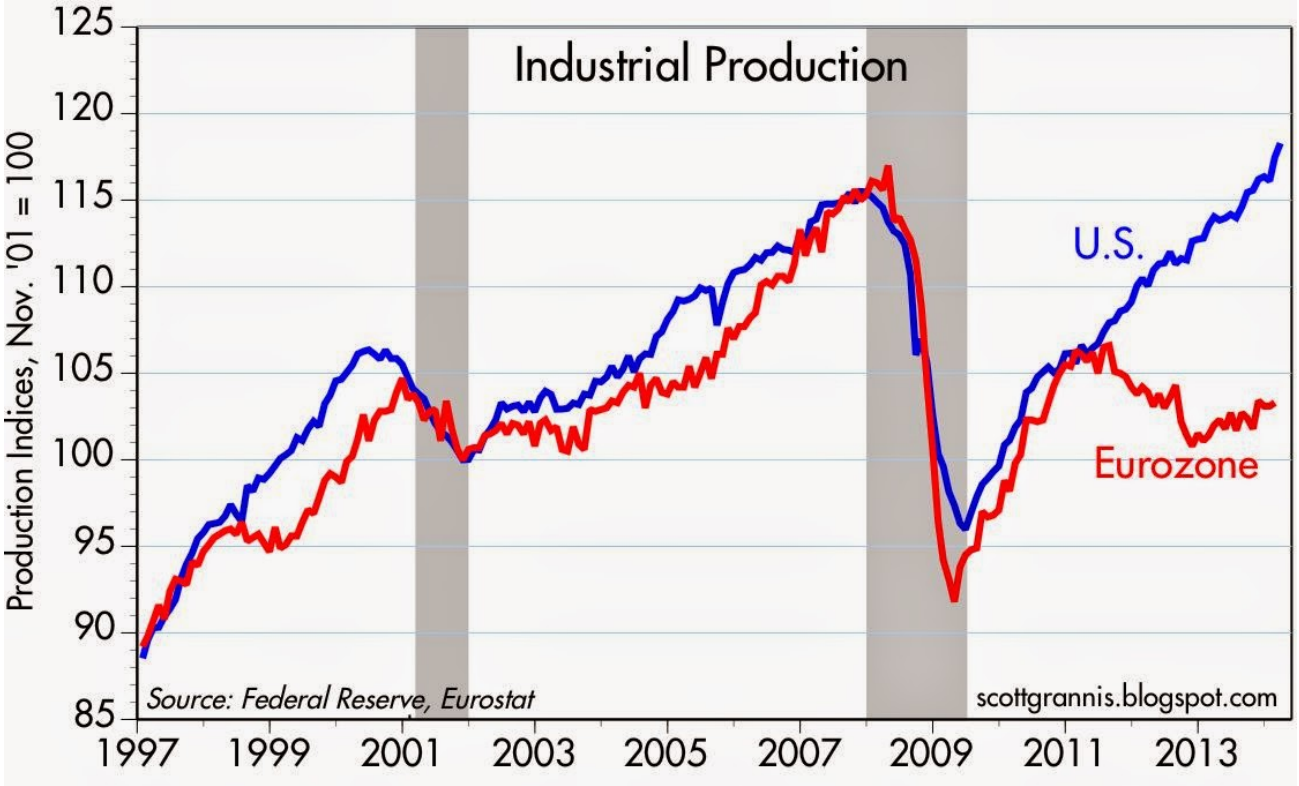
Regulated Outputs



Example:

- Regulated Output: Pitch angle (not measurable)
- Measured Output: Pitch rate (gyro)

The Economy: Feedback is **Dangerous**!



Prices are **FEEDBACK**

- Markets price goods by how much they are needed.
  - Scarcity and Demand to Price
- People produce goods based on how much money they can make.
  - Price and Cost of production to Production

Feedback Loops can be **DESTABILIZING**!

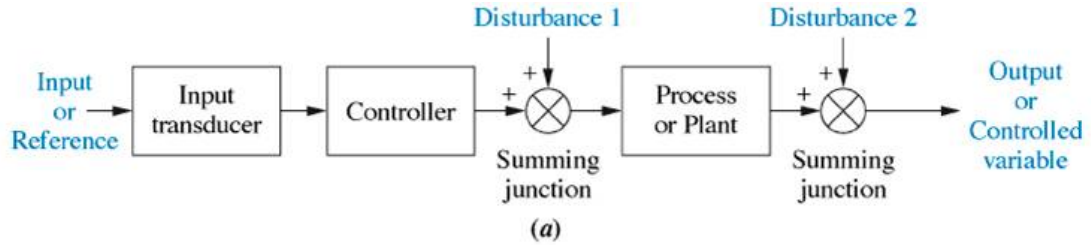
Block Diagrams and Control Systems

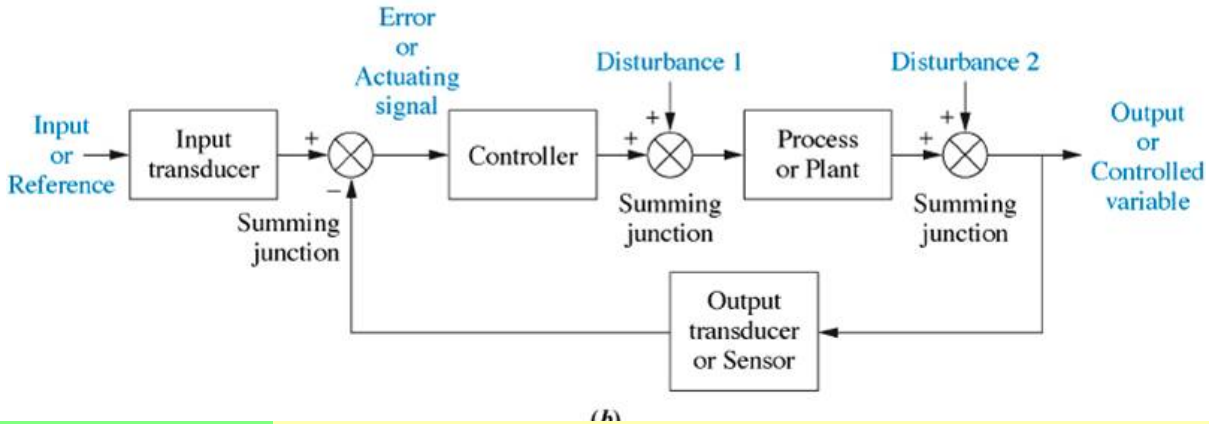
Definition 8.

Block Diagrams visually represent the flow of information between systems.

- **Systems** are represented using **Boxes**.
- **Signals** are represented using **Lines**.
- Simple addition and substraction are represented using **Circles**.

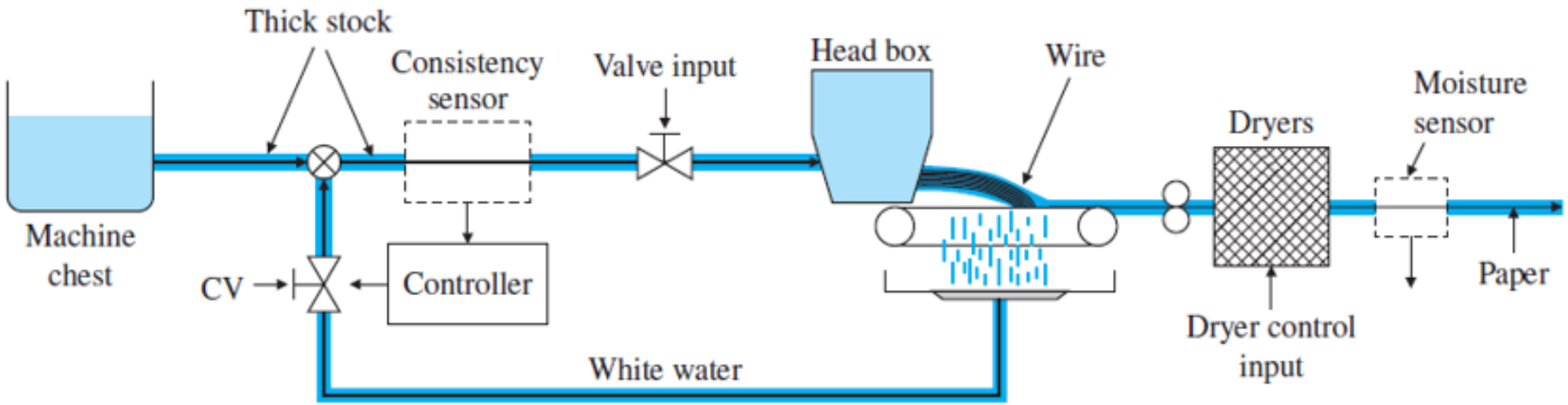
Illustration: Controllers can usually be represented as open-loop and closed-loop:





Example of a System

Paper Mill

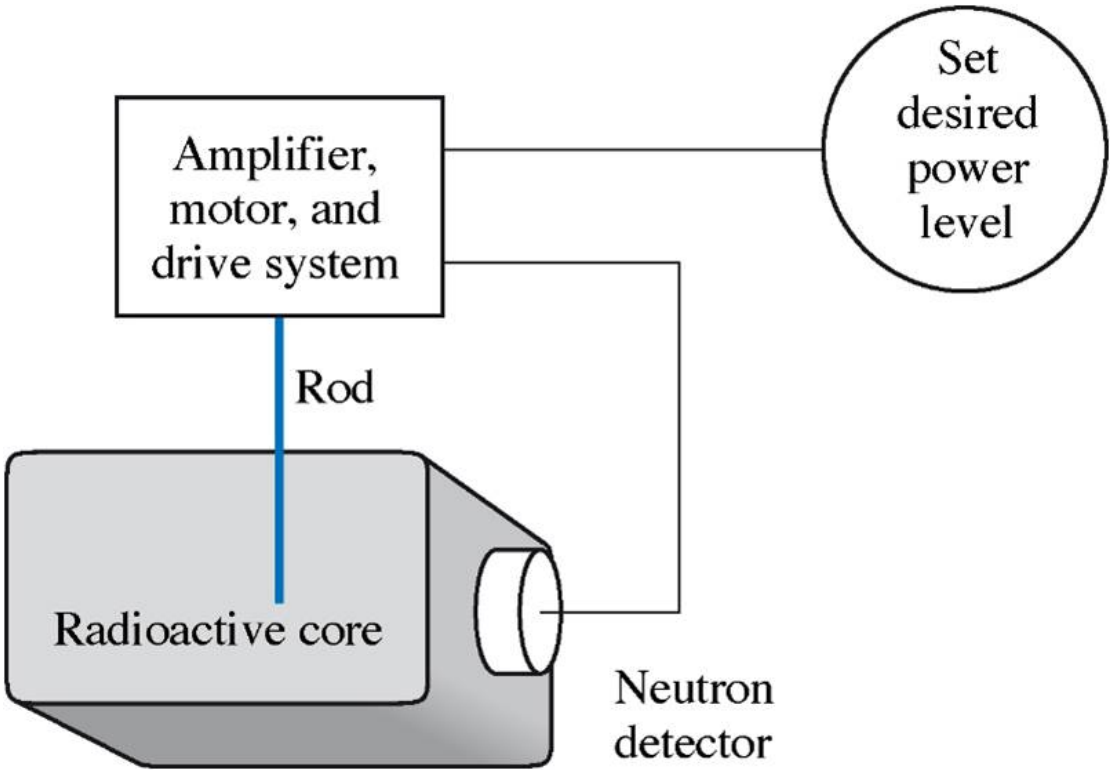


Things to look for:

- Dynamics
- Sensors
- Actuators
- Regulated Outputs
- Disturbances

Example of a System

Nuclear Reactor

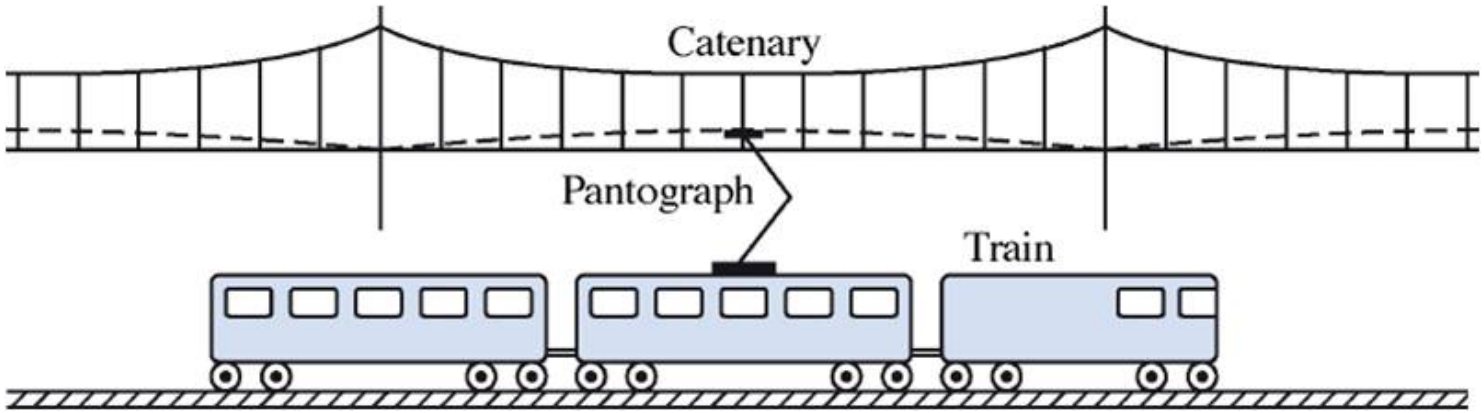


Things to look for:

- Dynamics
- Sensors
- Actuators
- Regulated Outputs
- Disturbances

Example of a System

Pantograph (High-Speed Electric Train)



Things to look for:

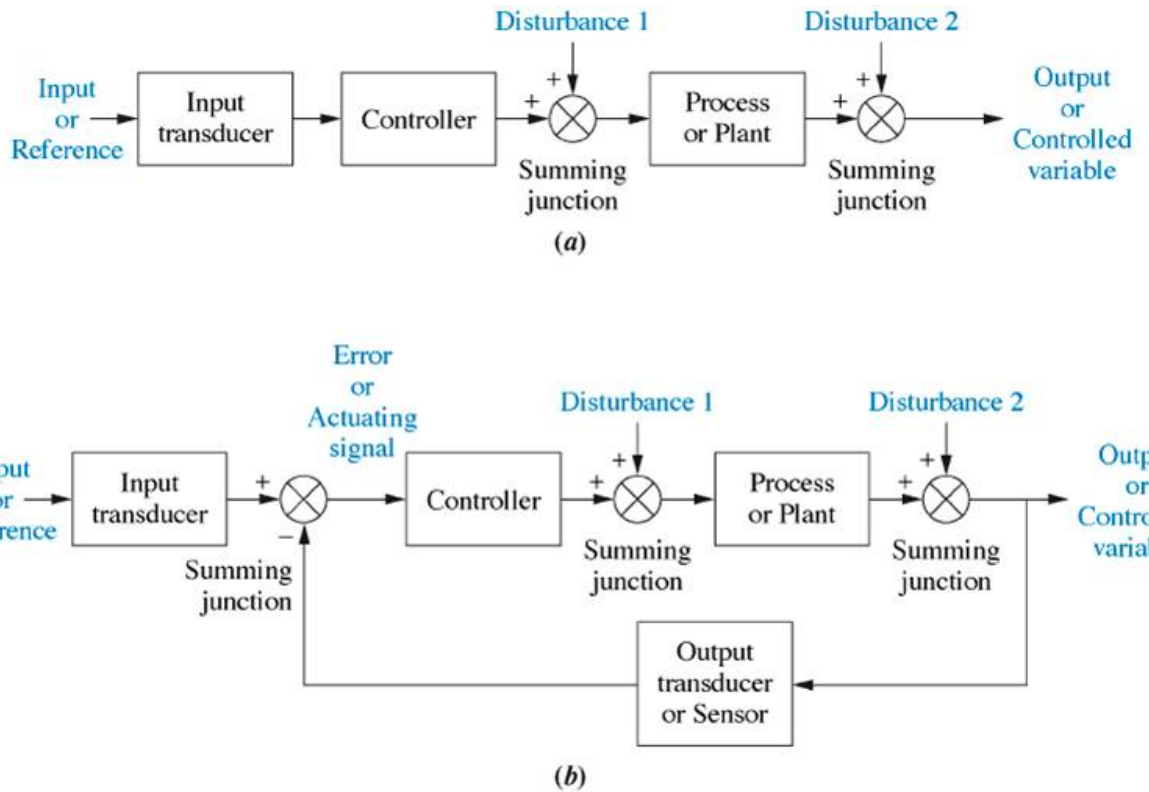
- Dynamics (Plant)
- Sensors (Measurable Outputs)
- Actuators (Controllable Inputs)
- Regulated Outputs (Objective?)
- Disturbances (Can't control)

Of Course there is also Open-Loop Control

Definition 9.

An **Open Loop Controller** has an actuator, but makes no measurements.

- No **FEEDBACK**
- No way to tell how well you are doing...



Examples of Open-Loop Control

- Choosing a Class
- A Pop-up Toaster
  - Actuators, Inputs, Outputs?
- Irrigation Systems

Problems?

Lets Compare Open- and Closed-Loop Driving

3D Google Maps Driving Simulator

2D Google Maps Driving Simulator

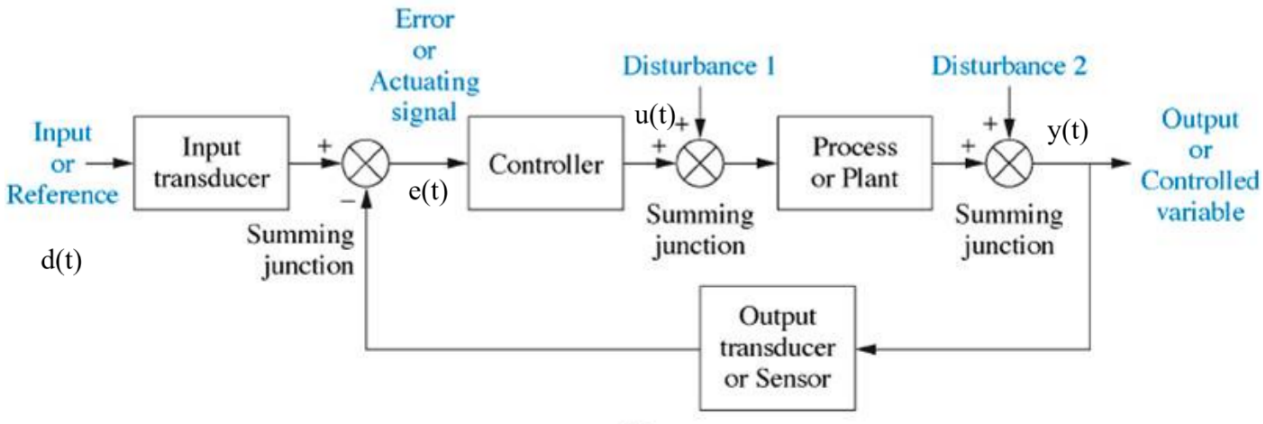
2D Driving Simulator without Google Maps

Control Concepts: Error Signal, Tracking, and Gains

Definition 10.

In the **Tracking Problem:** We want the output signal (measurement, $y(t)$) to match a desired input signal (disturbance, $d(t)$).

In a tracking problem, the **Error Signal** is the difference between the output signal and the desired input signal: $e(t) = y(t) - d(t)$.



Definition 11.

The simplest controller is a proportional **Gain (K):** Amplification of the error signal to feed back into the actuator — $u(t) = Ke(t) = K(y(t) - d(t))$.

Model-Based vs. Model-Independent Control

A Last Bit of Nomenclature

Open-Loop Control requires detailed knowledge of the System and Environment.

- How is this knowledge represented?

Definition 12.

A **Mathematical Model of a System** is any tool which allows us to *predict the output of the system* for any given input.

Examples: Differential Equations, Transfer Functions, Functions

- Comes from: Physical Principles; System Identification; Machine Learning

Definition 13.

A **Model-Based Controller** is a controller which uses a Mathematical Model of the system to map inputs to outputs (sensor signals to actuator signals).

Definition 14.

A **Model-Independent Controller** is a controller which is designed to work for ANY system.

History of Control: The Measurement of Time

Egyptian Water Clocks 1600BC

Significant in Commerce, Industry, Science, Medicine and Military

Time left is given by the amount of water left in the pot.
Problem: Measurement is limited to time left and by amount of water in pot.
Solution: Measure the amount of water that comes out of the pot.

History of Water Clocks

Time passed is amount of water in pot.

Problem: Water flow varies by amount of water in the top pot. ($\dot{x} = -\sqrt{x}$)

Solution: Maintain a constant water level in top pot.

History of Water Clocks

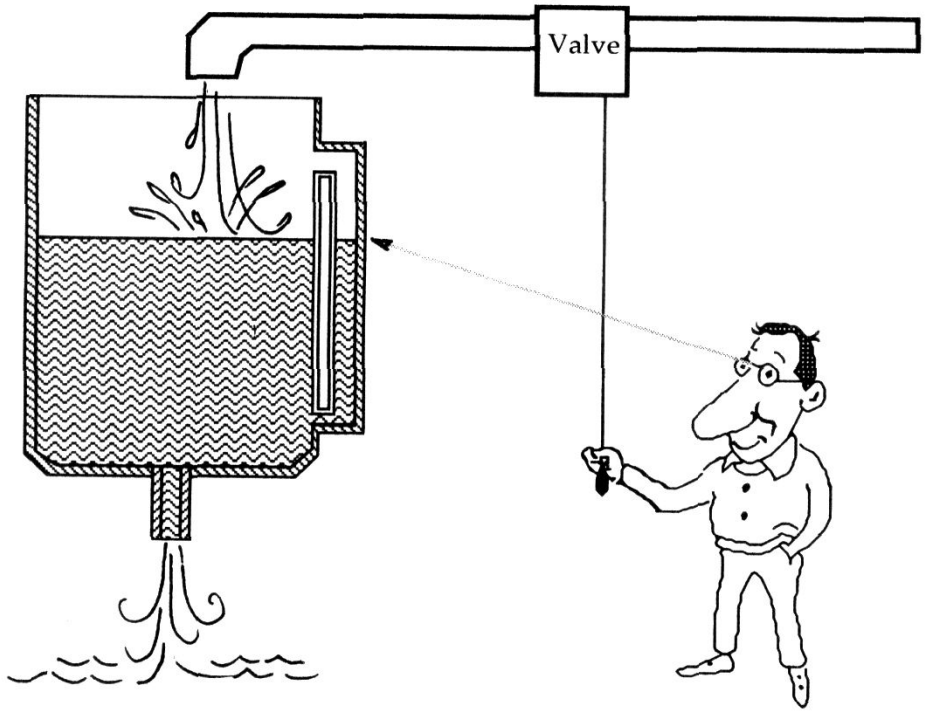


FIGURE 1.1. Level Control System. A sight tube and operator's eye form a sensor: a device which converts information into electrical signal.

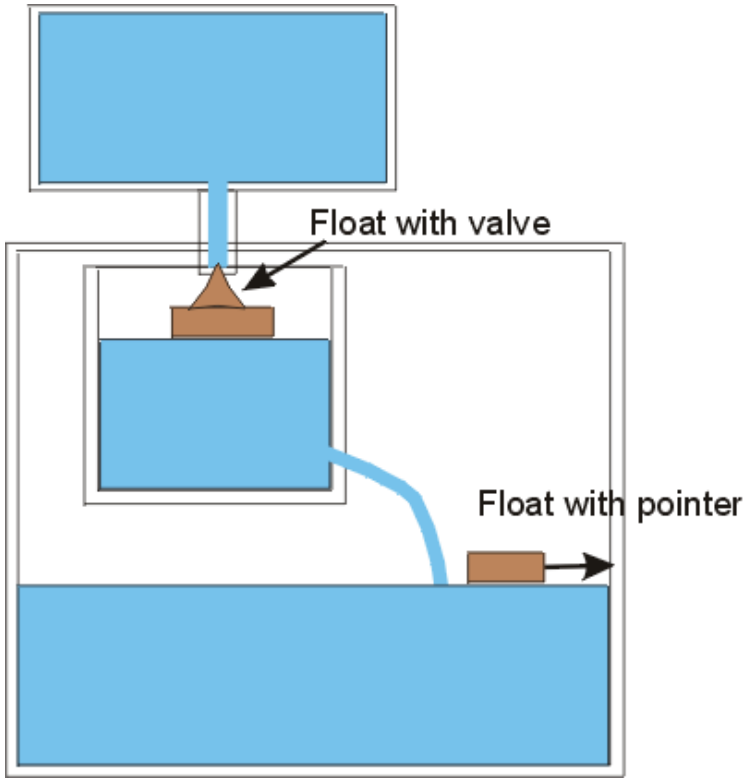
Problem: Manually refilling the top pot is labor intensive and inaccurate.
Solution: Design a control System (Inputs, Outputs?).

History of Water Clocks

Ctesibius c. 220-285 BC

Father of pneumatics

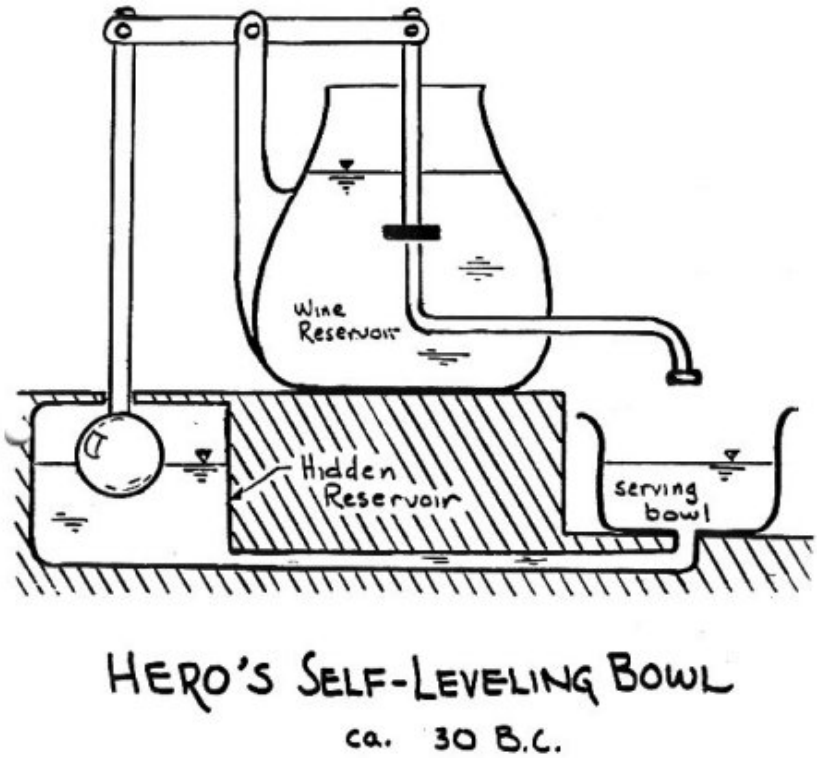
- Lived in Abject Poverty
- Created most accurate clock until Huygens (1657 AD)
- Overshadowed by better-known student Heron (Hero) of Alexandria

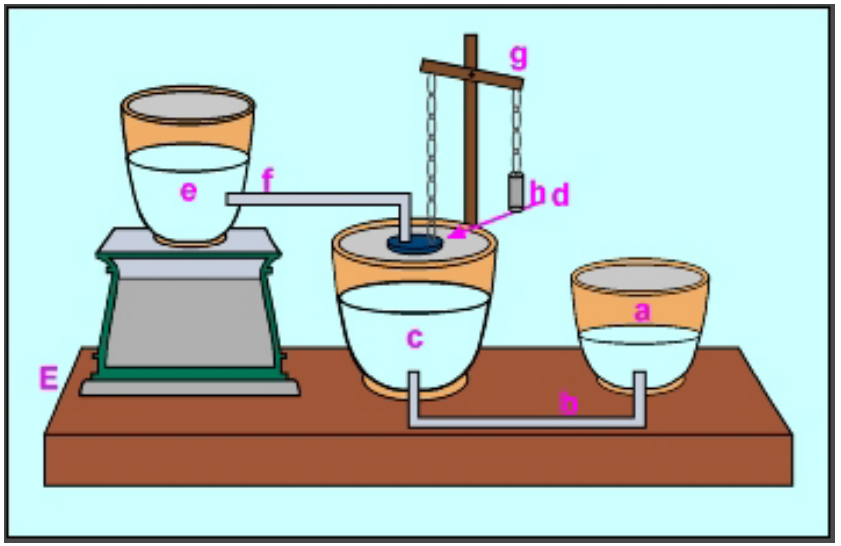


History of Water Clocks

Heron (Hero) of Alexandria c. 10 AD

As any good student, Hero used Ctesibius' water clock to perform party tricks.



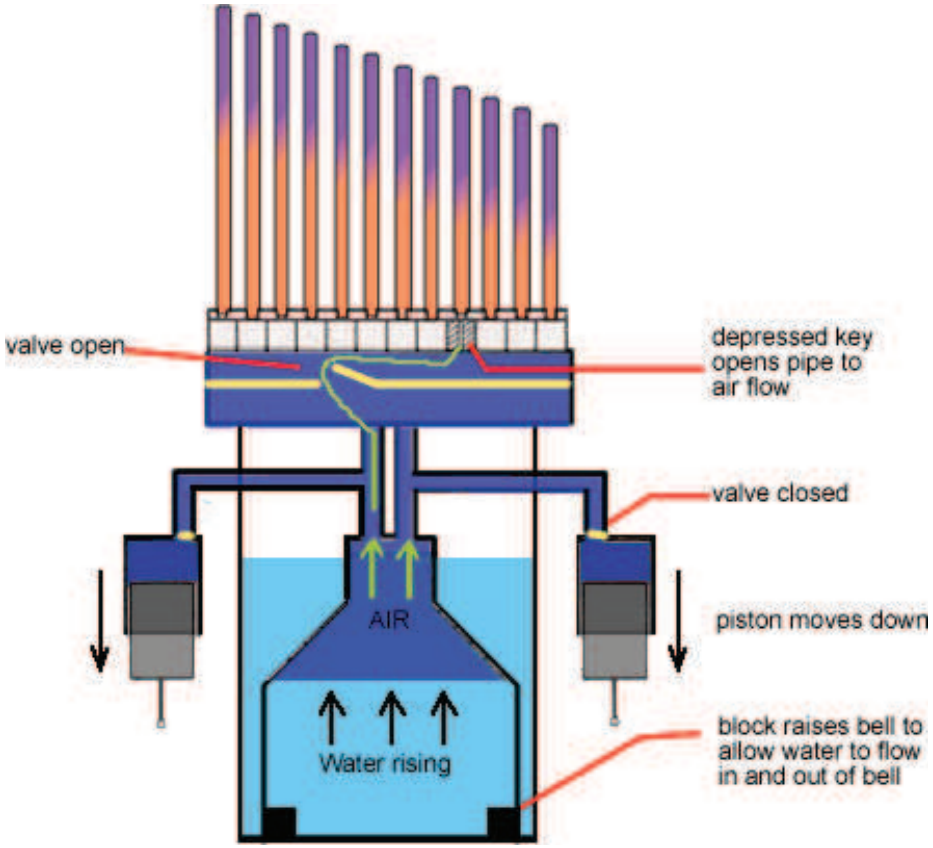


The self-replenishing wine bowl. (Inputs, Outputs?)

History of Water Clocks

The Pipe Organ

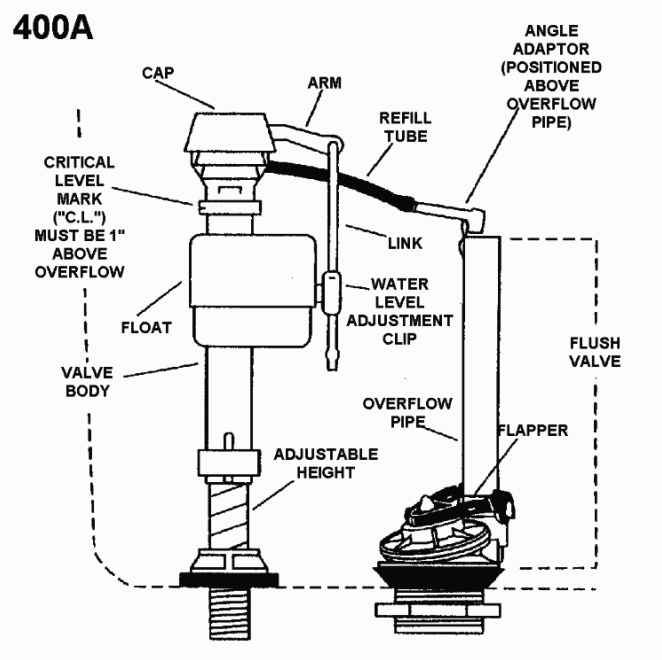
Ctesibius himself applied the principle of pneumatic control to create a pipe organ.



Maintains constant air pressure

History of Water Clocks

What do we use Ctsebius' water clock for today?



The Industrial Revolution

More Serious Applications

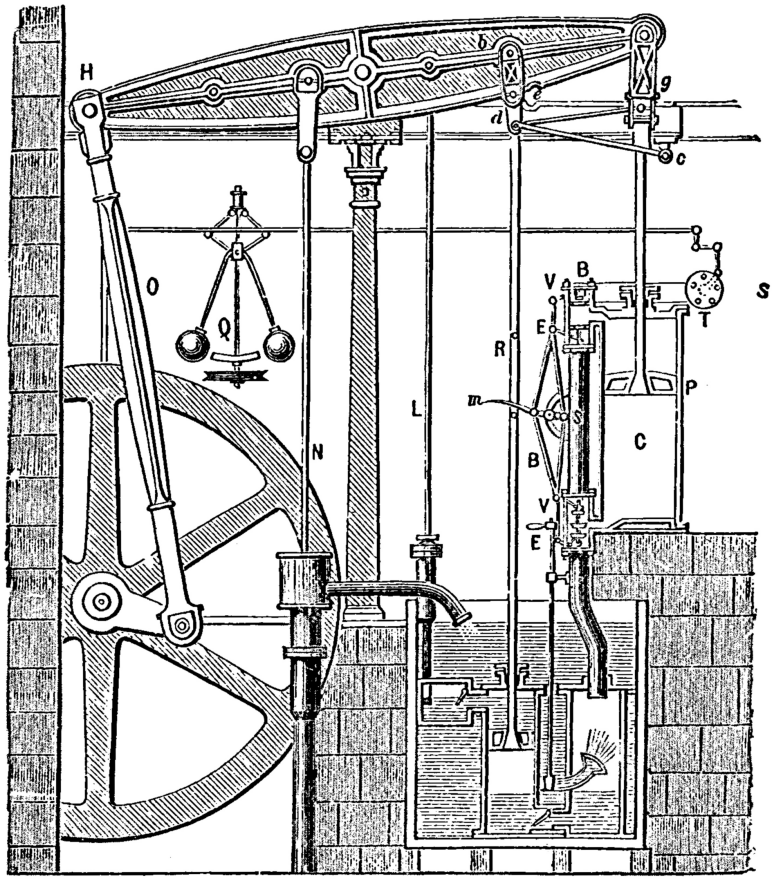
In addition to wine bowls, Heron also developed the steam engine.

Unfortunately, the Aeolipile was NOT CONTROLLED.

- No Feedback
- Missed chance for an industrial age?

Modern (Relatively) Steam Engines

James Watt and the Flyball Governor



Problem: Steam engines must rotate a piston at a *controlled speed*.
Solution: Control the pressure of steam.

The Flyball Governor (James Watt, 1736-1819)

Tracking Problem: Maintain a desired rate of rotation

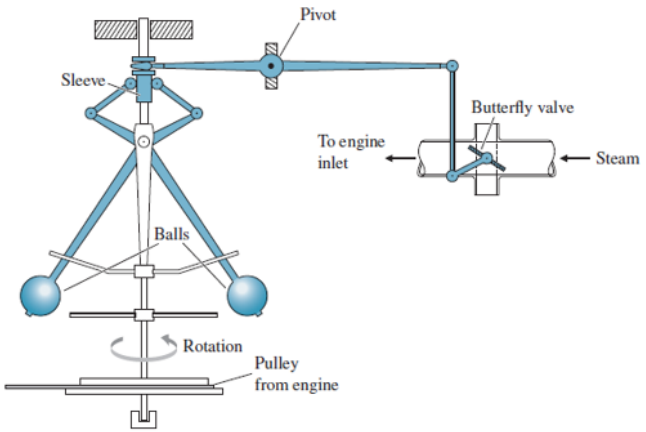
Sensor: Flyballs

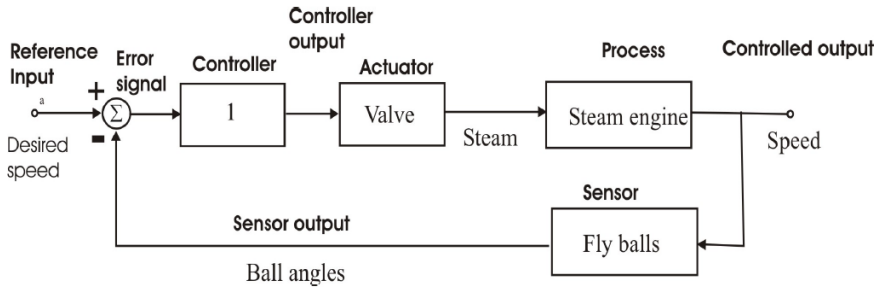
- Faster rotation $\Rightarrow$ More centrifugal force $\Rightarrow$ Angle of flyballs increase.

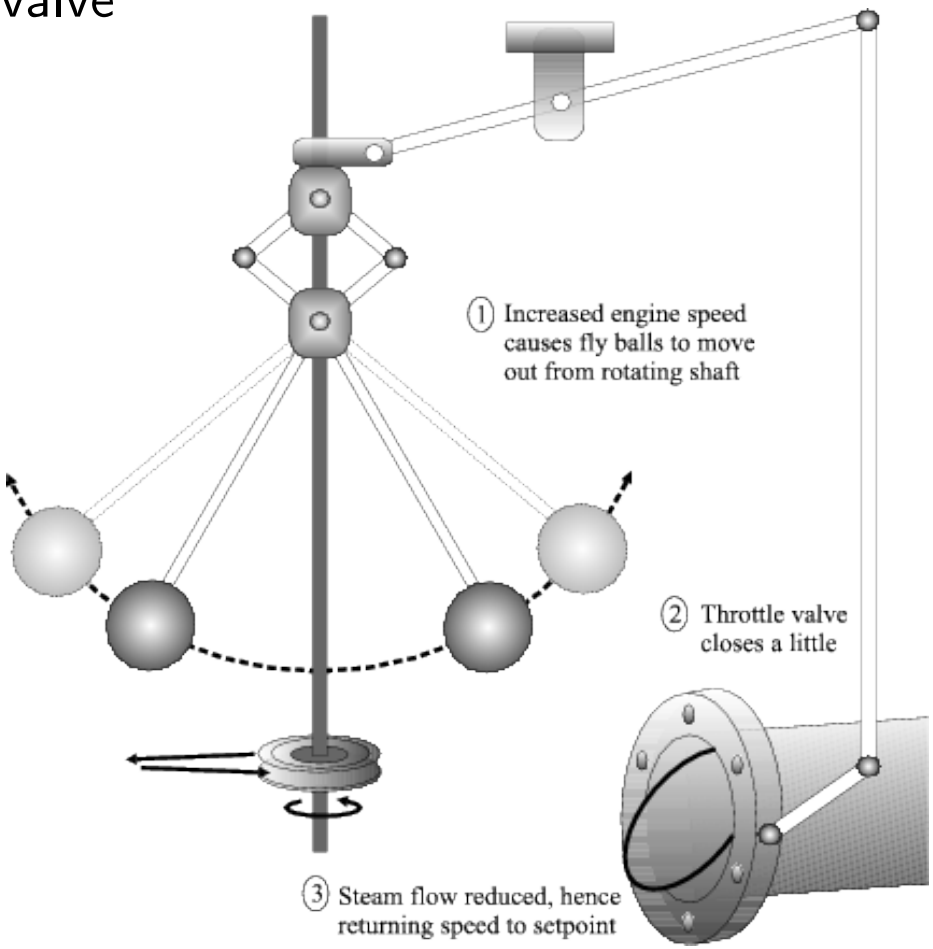
Actuator: Motion of pivot opens steam valve

Controller Gain:

- Ratio of pivot anchor lengths.





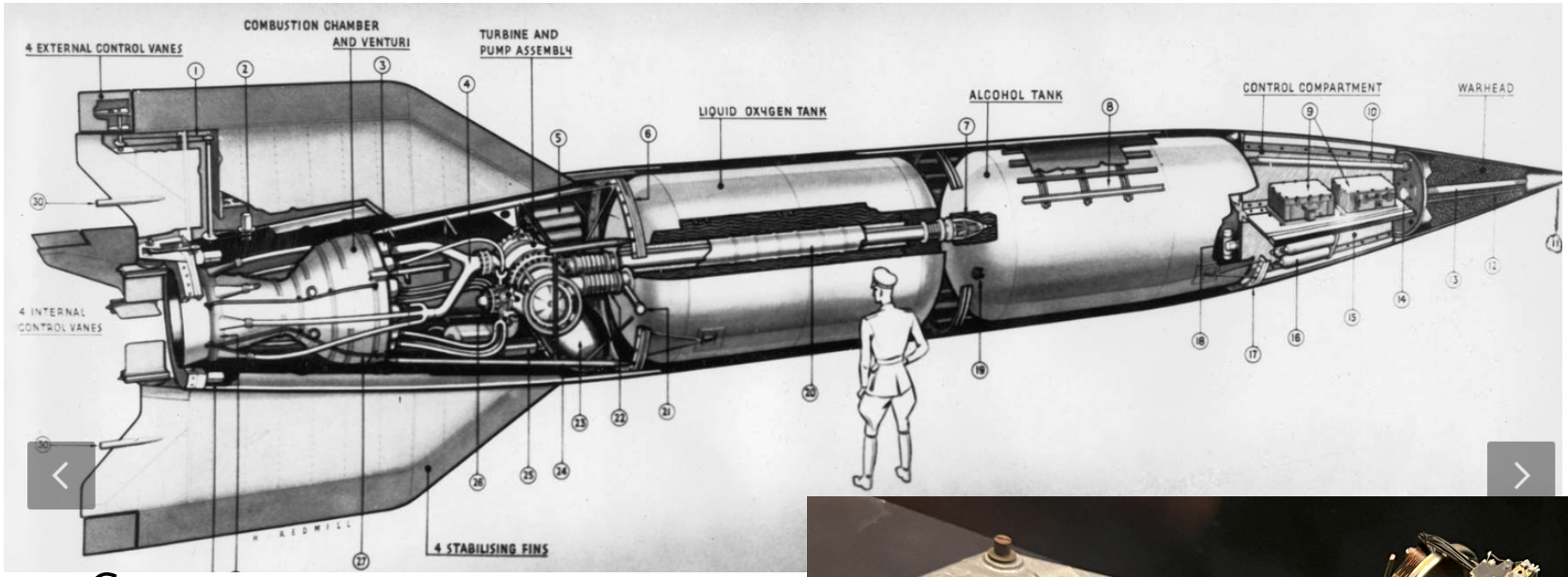


The Flyball Governor in Operation

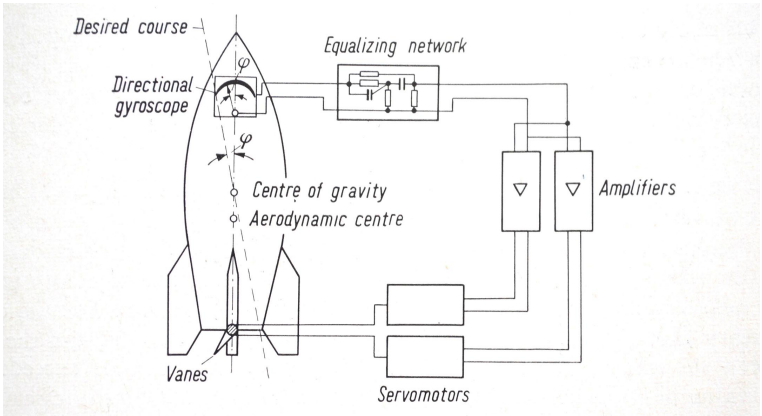
Stuart-Turner No9 Steam Engine

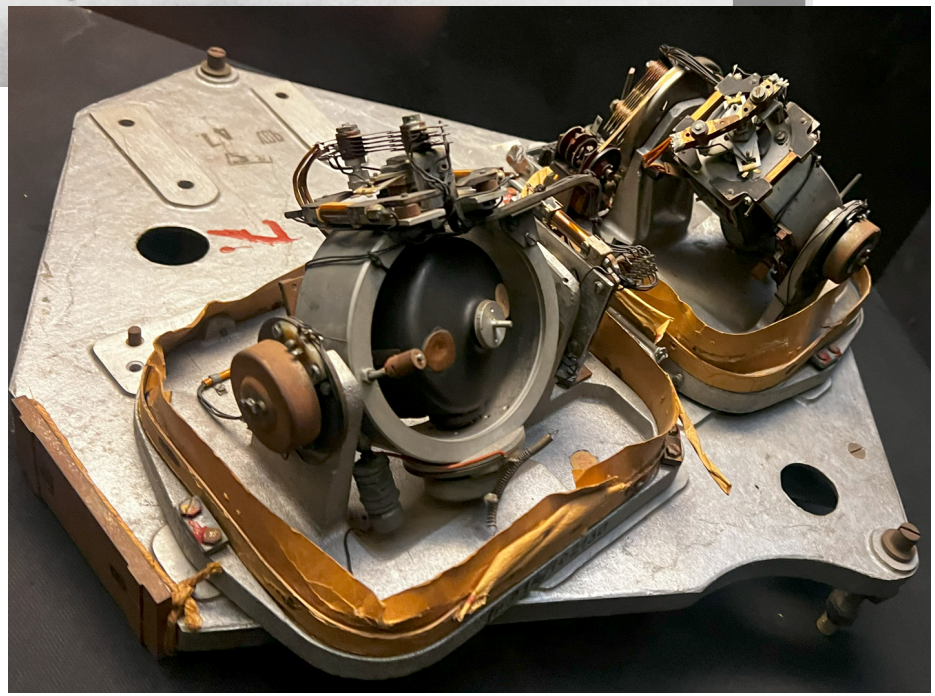
Rocketry (1940s and Wernher von Braun)

The first UAV was designed with a less than noble purpose.



Sensor: Gyrocompass
Actuator: Flaps and vanes
Regulation: Roll, Pitch and Yaw





Challenges for Control in the 21st century

Privatization of Space Travel

Challenges

- Safety
- Complexity
- Uncertainty

Links:

SpaceX Chopsticks Landing

Proton M launch Failure (Sensor installed upsidown)

Kepler Space Telescope (Actuator Failure)

Challenges for Control in the 21st century

UAVs and Drones (Delay, Sampled-Data)

Safe Interaction with

- Crowded Airspace
- Real-Time Obstacle Avoidance

Precision Control with

- Delayed Feedback
  $\dot{x}(t) = Ax(t) + Bu(t - \tau)$
- Lossy Connections
  $\dot{x}(t) = Ax(t) + Bu(t_k)$

Links:
X47 Drone Carrier Landing
Raff's TED talk

Challenges for Control in the 21st century

Self-Driving Vehicles

Challenges:

- Safety (Provable)
- Uncertainty (in model, environment)
- Other Drivers (Multi-Agent)
- Obstacles

Self-Driving Vehicles

- Google (Waymo)
- Tesla, Mobileye
- Toyota, Nutonomy

Links:
Waymo Navigation

Challenges for Control in the 21st century

Robotics

Control in Robots

- Complex Models
- Many actuators and sensors
- Interactions with the environment

Links:

Boston Dynamics, Atlas Mark 3

Tesla Robot

Control Design Process

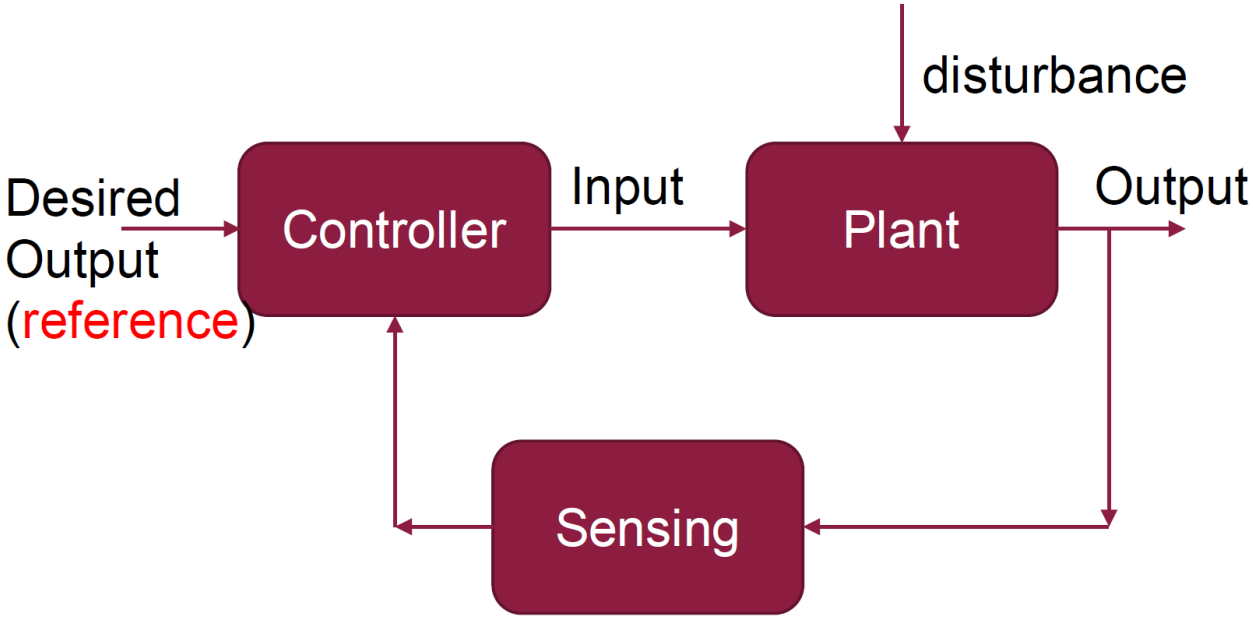
Step 1: Determine performance specifications

Step 2: Mathematical model

Step 3: Design controller in simulation

Step 4: Test in real life

Step 5: Repeat if needed



Overview of Course Objectives

Will that be on the final???

Part 1: System Analysis

- Given a system model:
  - Given an input, find the output
  - Predict the effect of Standard Inputs (Impulse, Step, Ramp, etc.)
  - Determine Stability
- Introduction to the Frequency Domain
- Given a desired response:
  - Determine the characteristic root locations

Part 2: Controller Design

- Given a system model:
  - Plot the effect of proportional gain (Root-Locus)
  - Plot the response to sinusoidal input (Bode Plot)
- Given a desired response:
  - Propose feedback controllers (PID and lead-lag) to achieve the desired response.